

Article

Digital Twin System Framework and Implementation for Grid-Integrated Electric Vehicles

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Abstract: Research on digital twins (DTs) in the power system field has mainly focused on implementing DTs for specific resources, while few studies on electric vehicle (EV)-based DT implementation have considered integration and interoperability between systems. This study introduces a DT-based EV system operation framework to address the aforementioned research gap. The framework implements individual EVs, charging stations, and charging station operators (CPOs) as DTs, enabling integrated operation with the power grid. The DT-based EV agent supports independent decision-making on power service participation by considering location information, distance, charging amount, spare time, and incentives. In addition, the CPO can establish an optimal incentive strategy to induce EV users to participate in grid power services. The proposed DT systems map information between EVs, charging stations, and the grid, enabling analysis and verification of the impact of participants on charging station operation, grid stability, and economic efficiency in an independent environment. The effectiveness and usability of the proposed framework were verified through a case study on an incentive-based demand response program.

Keywords: electric vehicles; EV behavior; digital twin system; grid integration; framework



Citation: Yu, J.-U.; Cho, K.-S.; Park, S.-W.; Son, S.-Y. Digital Twin System Framework and Implementation for Grid-Integrated Electric Vehicles. *Energies* **2024**, *17*, 6249. <https://doi.org/10.3390/en17246249>

Academic Editor: Chunhua Liu

Received: 4 November 2024

Revised: 2 December 2024

Accepted: 6 December 2024

Published: 11 December 2024



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1. Introduction

Electric vehicles (EVs) are becoming increasingly popular, as they offer several environmental benefits, including reduced carbon emissions, improved energy efficiency, and reduced air pollution [1]. Accordingly, efforts are being actively devoted to integrating EVs efficiently into the grid [2–4], and various technological approaches are being developed to secure the flexibility and stability of the power system.

Existing research on EV operations mainly focuses on technical and economic perspectives. References [5,6] focus on integrated vehicle-to-grid and grid-to-vehicle systems with EVs and renewable energy in microgrid (MG) environments. In Ref. [5], a planning model for EV charging stations considering a feed-in tariff and incentives is proposed, and [6] optimizes MG performance using a stochastic method. Reference [7] proposes battery degradation management strategies to secure the economic feasibility of EVs participating in the power system. Reference [8] discusses battery degradation modeling to support operation strategies. Recently, with the development of artificial intelligence, research has been conducted on EV charging forecasting [9,10] and dynamic energy scheduling [11] using learning-based algorithms, such as machine learning and reinforcement learning. These approaches demonstrate their ability to mitigate uncertainties in EV-based operations and effectively support dynamic decision-making.

A representative approach for supporting grid stability management is participation in ancillary services, such as voltage management, flexibility reserve, and frequency regulation using vehicle-to-grid technology. For instance, a method has been proposed to achieve grid load peak shaving and loss reduction through an EV smart charging model

for the distribution network in Stockholm, Sweden [12]. The study evaluated the annual loss, transformer congestion, and electricity bill changes through scenario-based simulation according to EV departure times. A voltage management method for distribution networks using EV resources has also been proposed [13]. The study utilized a real-time voltage regulation strategy based on model predictive control. In another study [14], a charging control optimization model was proposed for providing reserve power using EVs as resources; the model utilizes a frequency regulation and spinning reserve market participation strategy based on vehicle-to-grid technology. Ref. [15] proposed a method for participating in the frequency regulation market using EVs.

These studies focused on analyzing the effects of integrating EVs into the grid and establishing EV operation strategies from technical and economic perspectives. However, these strategies do not reflect the actual operating environment accurately because they do not sufficiently consider user behavior, which affects EV mobility, and the variability of charging patterns. Additionally, limited simulations were performed because simple and generalized models, the data used, and the construction of case study scenarios were complicated. Agent-based models have been proposed to simulate the driving and charging behaviors of EVs to address this issue [16–19]. Specifically, researchers have proposed advanced modeling techniques based on the responsiveness of EV users to their driving and charging behavior. Although this approach is effective for understanding user behavior and establishing operation strategies, it has limitations in accounting for the complex interactions between other systems in a smart grid environment.

To overcome these limitations, digital twin (DT) systems have been explored [20,21]. DT systems represent a replica of a physical system in a virtual space, providing an environment for real-time monitoring, forecasting, operation, and analysis. They can also be used to observe the interactions between systems constituting a complex smart grid environment in more realistic detail. Recent research in the power system field has focused on utilizing DTs for the operating and monitoring of various energy systems, such as distribution systems and distributed resource operations. For instance, the DT system of [22] has been proposed as a comprehensive anomaly detection and optimization solution for hydroelectric power plant systems. In Ref. [23], a DT framework and energy management strategy combined with the Internet of Things and renewable energy modeling was proposed for hybrid microgrids. A framework utilizing an engineering approach to introduce DT technology into grid operation has also been presented [24]. A holomorphic embedding-based model of multiple energy networks accommodating electricity, gas, and heat has been developed, as well [25]; this strategy utilizes a DT architecture for multiple energy networks and focuses on mechanism-based modeling. A DT framework for simulating distributed energy resources (DERs) in distribution systems has been proposed [26]; this framework comprises four core modules—DER, a distribution system, an energy management system, and an electricity consumer—and analyzes the impact of grid interconnection. In Ref. [27], a hierarchical DT architecture supporting grid planning and operation has been introduced; this architecture emulates the actual grid conditions within a real-time simulation and verifies the time delay between the physical grid and its DT. Another DT has been implemented to improve the efficiency of renewable energy sources [28].

These studies focused on implementing DT for specific resources and systems. When implementing a DT for EV resources, researchers often emphasize the accuracy of individual models. However, EVs should be considered a part of the entire system from a power service operation perspective because of the complex interactions among the large number of EVs, charging points (CPs), charging station operators (CPOs), and distribution systems.

This paper proposes an EV system operation simulation framework based on an advanced DT model that can be linked with the operation simulation of a distribution system. This framework enables the creation and management of DTs for individual EVs, CPs, and CPOs and integrates these elements with grid operations. In the proposed model, DT-based EVs independently make decisions to participate in power services by considering parameters such as driving distance, location, battery status, and spare time,

as well as external factors, such as incentives for service based on an advanced elasticity model. Considering the responses of the EV users, the impact on distribution system stability, operation costs, and stakeholder economics can be analyzed and verified through the proposed framework.

The contributions of this study to the existing literature are as follows:

- A DT-based e-mobility simulation framework integrated with a distribution system is proposed.
- An interaction model that operates independently among EV users, CPOs, and the distribution system is developed.
- An independent decision-making model for EVs based on an elasticity model, considering the driving distance, incentives, charging, and time pressure, is applied.

The analysis of various mutual influences, such as distribution system stability, operational costs, and user utility, is supported based on EV user responses.

The remainder of this paper is organized as follows. Section 2 introduces the simulation platform environment where the proposed DT system for EVs is applied. Section 3 describes the e-mobility framework. Section 4 explains the modeling of the main function of the EV-DT model. Section 5 presents a case study and discussion. Finally, Section 6 presents the conclusions of this study.

2. Simulation Platform for Grid-Integrated EV Operations

This section introduces the simulation platform where the proposed DT-based EV framework is implemented. Smart Grid Simulator (SGSim) is an integrated DT simulation platform that can simulate a smart grid environment and is reliable across various power system applications [29–31]. The platform is designed with a multi-agent-based modular structure, where smart grid components, such as load customers, DERs, and the power system, operate independently. This enables easy addition or modification of various smart grid components through modularization, offering advantages in terms of scalability and flexibility. Additionally, it supports various smart grid agent functions, enabling the analysis of integrated operations and interactions between systems, similar to real-world environments.

In SGSim, a group of agents for a specific simulation is defined as a “Package”. Figure 1 shows the class diagram of SGSim. Here, a common package manages the overall simulation of the DT, including each agent’s execution order, the message flow coordination, the interaction between agents, and event handling. The customer package enables independent operational simulations of load customers and DERs and also supports data generation and forecasting simulations for these components. The grid package provides simulation functions for both normal and fault states of the distribution system and can be used to conduct grid impact analysis through integration with other application systems. Based on this environment, two DT packages supporting simulations for peer-to-peer operation and virtual power plant operation have been developed in previous studies [30,31].

In this study, a DT-based e-mobility package framework of [31] was expanded. This framework comprises a DT-based EV agent, a CP agent, and a CPO agent, enabling integrated operation with the grid system. Section 3 details the framework for the e-mobility package.

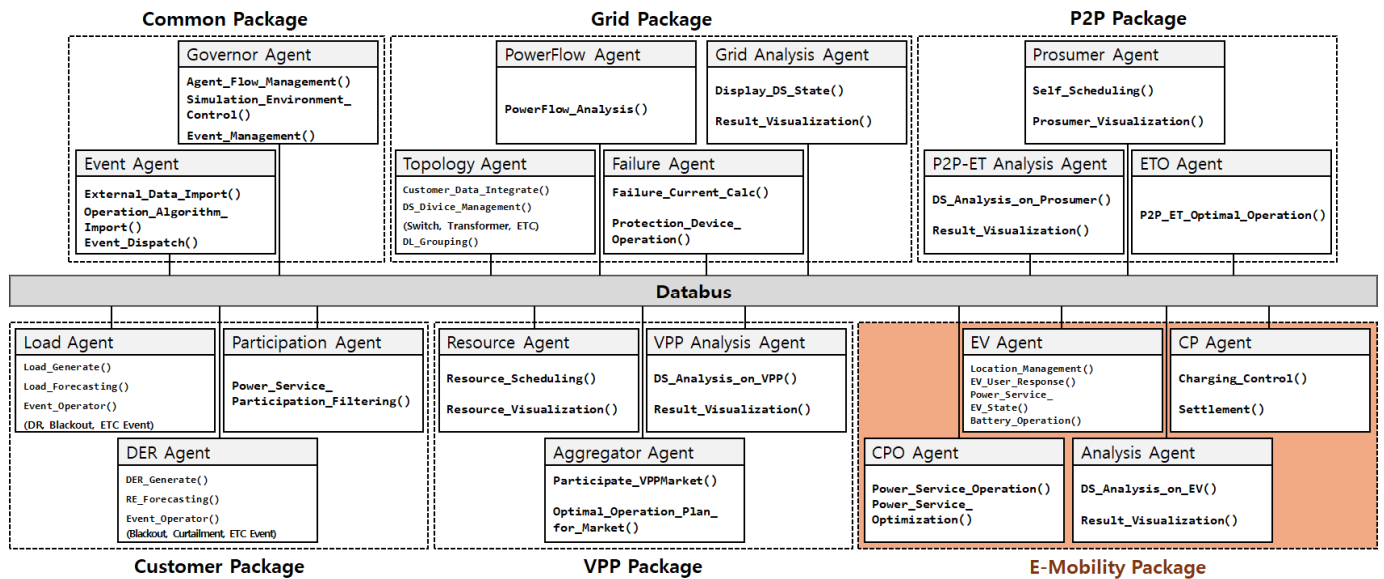


Figure 1. Class diagram of SGSim.

3. E-Mobility Package Framework

3.1. Mobility Package Architecture

The DT-based e-mobility package comprises four main agents: the EV agent, CP agent, CPO agent, and analysis agent. The EV agent supports independent decision-making simulation for participation in power services by considering the behavioral dynamics of EV users. The CP agent supports individual charging and discharging simulations of charging stations, and the CPO agent supports simulations for establishing strategies to participate in power services, such as voltage management and frequency control.

Each agent comprises modules that perform specific functions, and the agents within the package can share and access data in real time via the databus. This allows the DT-based e-mobility package to interact with the grid package, supporting integrated grid operations. Specifically, the EV and CP agents interact with the grid package's topology agent and power flow agent to share the location and status information of EVs and chargers. The grid package connects with the load agent and DER agent from the customer package to configure loads and DERs within the distribution system. Information and signals for power service operation are managed through the event module of the common agent, which interacts with the power service operation module of the CPO agent. Figure 2 shows the e-mobility package architecture, highlighting the interactions between the agents within the framework. Section 3.2 explains each agent's functions and their role in modeling complex system interactions. Section 3.3 focuses on enabling the integrated operation of EVs and energy services.

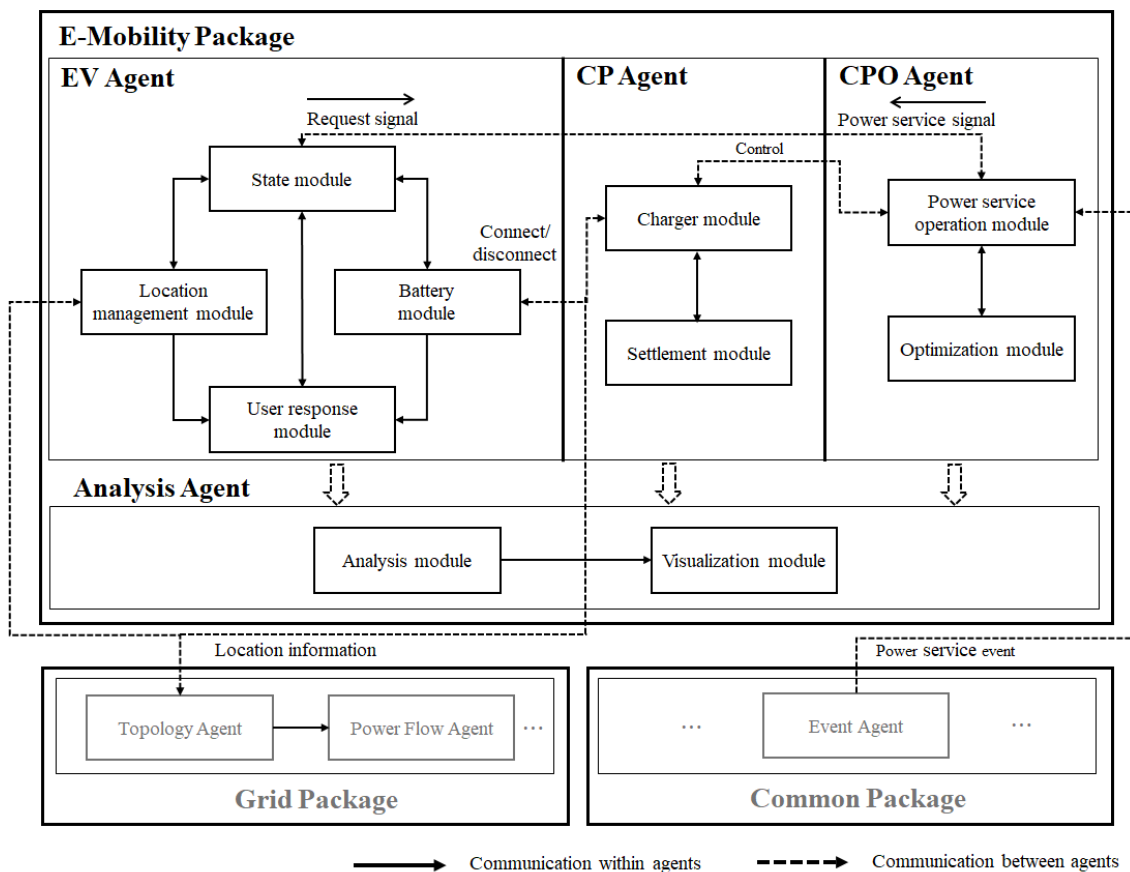


Figure 2. E-mobility package architecture.

(A) EV Agent

The EV agent independently makes decisions about participating in power services based on details such as EV driving, location, distance, charging, and spare time, as well as external factors, such as incentives. The EV agent comprises four modules, and the interactions among these modules are shown in Figure 3. The main functions of each module are as follows:

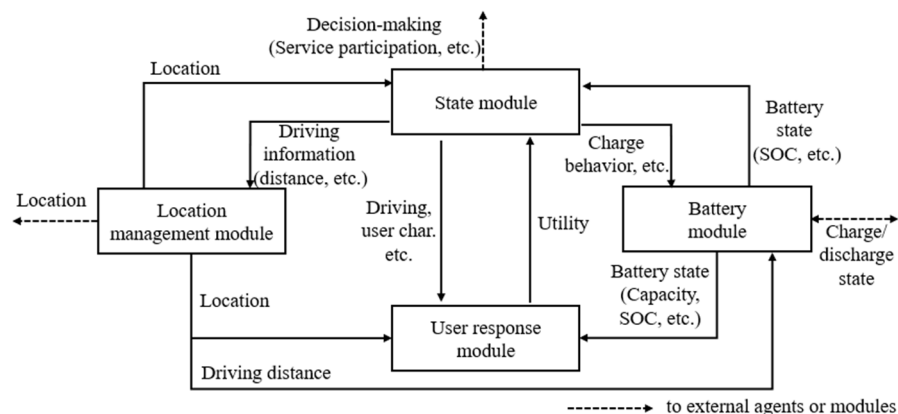


Figure 3. EV agent architecture.

3.2. Functions of E-Mobility Package

- The state module simulates the behavior of EV users. Specifically, it supports independent EV user decision-making regarding activities such as driving, charging, and participating in power services by interacting with other modules within the EV agent. The EV user decision-making model is described in detail in Section 4.1.

- The user response module quantifies the utility of individual EV users based on details such as location data, distance to CPs, battery state, spare time, charging price information, and incentives. It interfaces with the location management module, battery module, and state module to obtain the information required to calculate the utility. The quantified results are used as key indicators for decision-making in the state module. The operation model is described in detail in Section 4.2.
- The location management module manages the spatial and temporal positioning of individual EVs. It is linked to the battery module to support the calculation of battery consumption according to driving distance and is linked to the topology agent of the grid package to facilitate location information mapping for individual EVs based on node units in the power distribution system.
- The battery module interacts with the state module, the location module, and the CP agent's charger module to simulate the battery status (state of charge (SOC), degradation, etc.) according to the driving, charging, and discharging details.

(B) CP Agent

The CP agent supports simulations at the individual charger level within CPs and performs functions such as charging control and settlement. It also supports control simulation of operational strategies for power services involving CPO agents. The CP agent comprises a charger module and a settlement module, the main functions of which are outlined as follows:

- The charger module supports the simulation of a CP's charger operation. It interacts with the battery module of the EV agent to provide charging power according to the charging behavior of EV users or supports charger control simulation according to the operation strategy by linking with the power service operation module of the CPO agent. Additionally, it is linked with the topology agent of the grid package to provide location and charging/discharging power information for grid simulation.
- The settlement module performs settlements related to the charging and participation of EVs in power services. Additionally, it reflects changes in the CP's charging prices and the incentive strategies related to participation in power services from the CPO.

(C) CPO Agent

The CPO agent, which comprises a power service operation module and an optimization module, supports the simulation of charging service and power service operations. The main functions of each module are as follows:

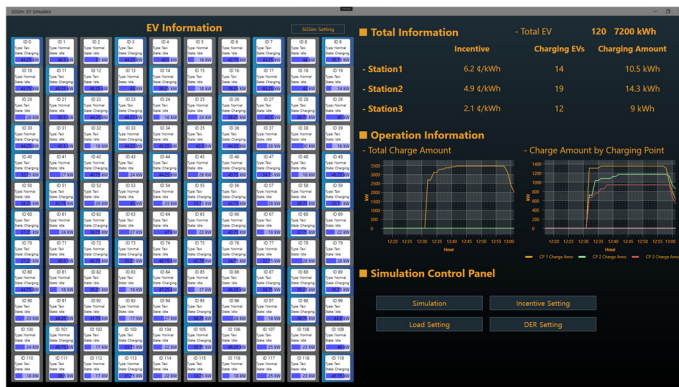
- The power service operation module manages the overall power service operation and interacts with the EV agent's state module and the event agent of the common package. Additionally, it considers operating strategies received from the optimization module to provide power service signals to EV users or charging/discharging control signals to the charger module.
- The optimization module establishes the optimal operation strategy for the power service. The power service information is provided by the power service operation module and feeds back optimization results. The optimization model is described in detail in Section 4.3.

(D) Analysis Agent

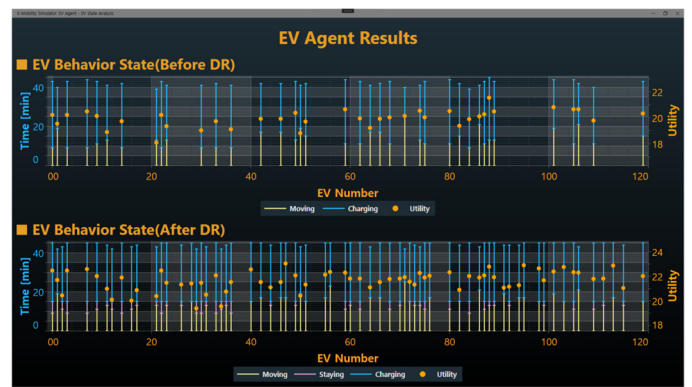
The analysis agent analyzes the main simulation results from the e-mobility package and provides real-time monitoring and visualization functions. It comprises an analysis module and a visualization module, the main functions of which are as follows:

- The analysis module filters, sorts, and aggregates the simulation results of each agent of the e-mobility package and grid package in real time using language-integrated queries.
- The visualization module visualizes the data analyzed in the analysis module. In the grid package, a dashboard and user interface (UI) are designed using extensible application markup language within Windows Presentation Foundation. Additionally, to synchronize data with the UI, the Model–View–ViewModel pattern was applied.

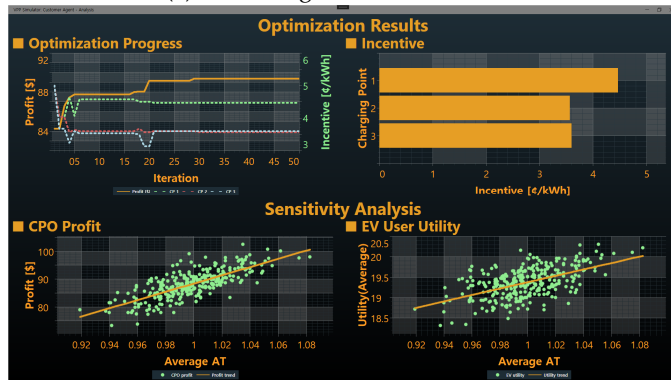
- Figure 4 shows the main visualization UI/UX of the e-mobility package. Figure 4a shows the comprehensive results of the EV package simulation, including incentives, EV participation information, and charging amount for each CP according to the power service operation. Figure 4b shows the visualization results of the EV agent. This visualization result can determine the changes in EV battery and user states before and after participating in power service, and includes staying, moving, and charging states of individual EVs, battery SOC, and user utility information. Figure 4c shows the results of the CPO agent, including the optimal incentive calculation process and results, and the sensitivity analysis results of key parameters. Figure 4d shows the visualization results of the grid package. The package can determine the change in the distribution system states according to the power service operation of CPO.



(a) EV Package simulation results



(b) Analysis of EV and CP agent results



(c) Analysis of CPO agent results



(d) Analysis of distribution system

Figure 4. Visualization of analysis agent.

3.3. Simulation Flow of E-Mobility

The simulation flow of the e-mobility package and its interaction with the grid package are depicted in Figure 5. The simulation flow is as follows:

(A) Normal Operation

- Operation ① is a simulation flow for a normal operation situation, which does not include any power service request. When the simulation starts, the EV agent's state module updates the EV's spatiotemporal information and status (charger connection, charging requirement, driving status, etc.) based on the interactions between the modules in the EV agent. The EV state is updated from the EV agent at each simulation step.
- Subsequently, the battery module is mapped to the charger module of the CP agent based on the charging connection information. If the EV is driving, its location information is shared with the topology agent of the grid package. Then, the location information from the charger module is updated to this topology agent, which completes the spatiotemporal mapping of the EV and charger.

- The charger module charges the battery based on the EV user's charging requirement. Then, the charging state information is sent to the CPO agent and the power flow agent of the grid package. Thereafter, the power flow agent performs a power flow analysis by updating the charger state (charging power) and the information from the topology agent.

(B) Power Service Operation

- When power service signal ② is transmitted to the CPO from the event module during the simulation, power service operation ③ is simulated.
- The CPO establishes an optimal operation strategy based on the power service information. In this study, the power service targeted incentive-based demand response (DR) to resolve overvoltage situations in the distribution system. The optimal operation strategy for the CPO is to calculate the optimal incentives to induce EV charging. The optimal operation model is explained in Section 4.

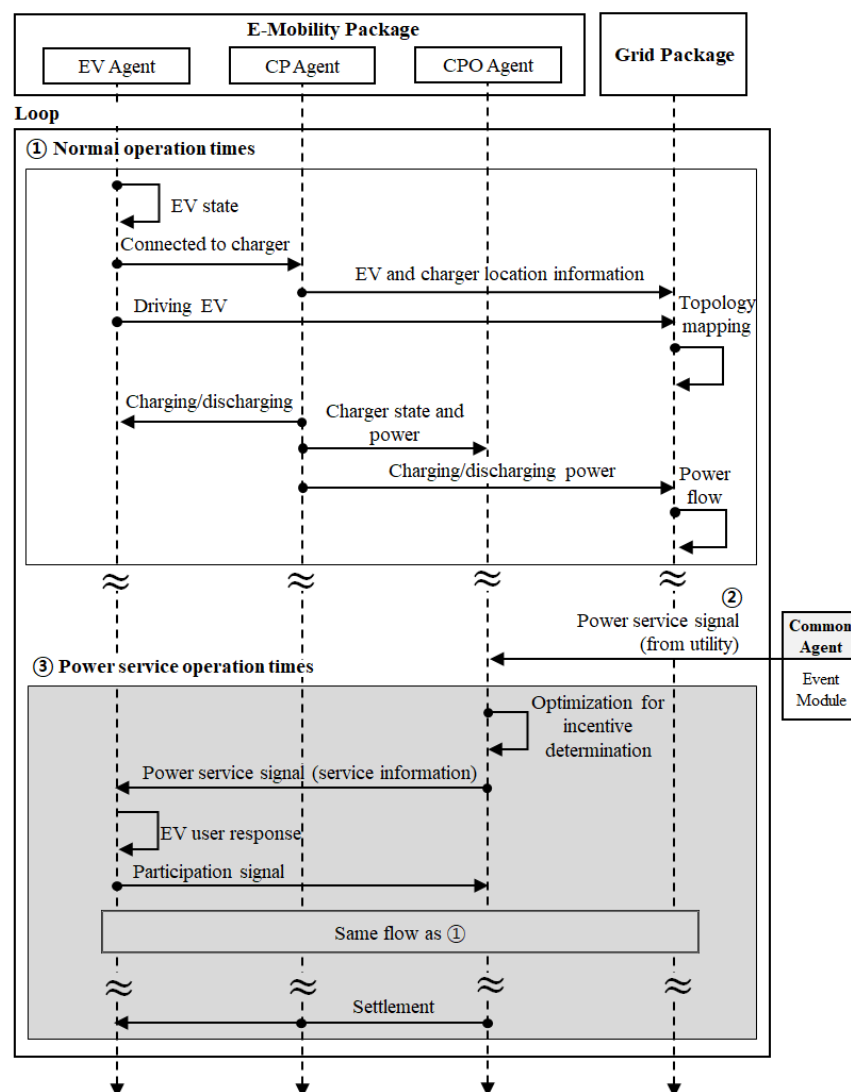


Figure 5. Simulation flow of e-mobility package with grid package.

- After the optimal incentive for the CPO is determined, the power service information, including incentives, is transmitted to the EV agent's state module. Based on the EV state information, including incentives, the EV user's responsiveness is calculated in the EV agent's response module, and the EV state, including the decision on whether to participate in the power service, is updated.

- Thereafter, the information regarding power service participation is transmitted to the CPO's power service operation module, and the subsequent real-time operation sequence is the same as ①.
- When the power service ends, a settlement is performed in the EV agent's settlement module, and the settlement result is delivered to the individual EVs that participated in the service.

4. E-Mobility Function Models

4.1. EV State Model of EV Agent

An EV user selects ($u_{t,c}^*$) a CP that maximizes their utility by participating in the power service, which can be expressed as Equations (1) and (2) [32]. The utility σ_t is defined as the energy that the EV user wants to charge. The utility model is described in detail in Section 4.2.

Based on the rated charge (P_c^{cp}), the charging energy ($E_{t,c}$) of the EV at time slot t can be expressed as Equation (3), while Equation (4) represents the total charging energy ($E_{t,c}^t$) during participation in the power service. Equations (5)–(7) represent the charging start time (τ_c^{st}), time spent at the CP for charging (τ_c^{cs}), and the charging end time (τ_c^{ce}) for the EV, respectively [32]. Equation (8) represents the charging state of the EV at a CP, which is determined at a time between τ_c^{cs} and $\min(\tau_c^{pe}, \tau_c^{ce})$. Equation (9) represents the SOC constraint for the EV battery charging.

$$u_{t,c}^* = \arg \max_{u_{t,c}} \sigma_c^t \quad (1)$$

$$\sigma_c^t = \sum_{t \in PT} u_{t,c} \cdot \sigma_{t,c}, \forall c \in CT \quad (2)$$

$$E_{t,c} = \min(u_{t,c} \cdot \sigma_{t,c}, \varepsilon \cdot u_{t,c} P_c^{cp} \cdot \eta^c), \forall t \in PT, \forall c \in CT \quad (3)$$

$$E_{t,c}^t = \sum_{t=\tau_c^{cs}}^{\min(\tau_c^{pe}, \tau_c^{ce})} E_{t,c}, \forall t, \tau_c^{cs}, \tau_c^{de} \in T, \forall c \in CT \quad (4)$$

$$\tau_c^{cs} = \max(\tau_c^{ps}, \tau_c^{ar}), \forall c \in CT \quad (5)$$

$$\tau_c^{st} = \frac{E_{t,c}^t}{\varepsilon \cdot P_c^{cp} \cdot \eta^c}, \forall c \in CT \quad (6)$$

$$\tau_c^{ce} = \tau_c^{cs} + \tau_c^{st}, \forall c \in CT \quad (7)$$

$$u_{t,c} = \begin{cases} 1, & \tau_c^{cs} \leq t \leq \min(\tau_c^{pe}, \tau_c^{ce}) \\ 0, & \text{otherwise} \end{cases}, \forall t \in T, \forall c \in CT \quad (8)$$

$$s_{t,c}^{min} \leq s_c^{it} - \varepsilon \cdot P_{t,c}^{evm} / \eta^d + \sum_{g=1}^t E_{t,c} \leq s_{t,c}^{max}, \forall t \in T, \forall c \in CT \quad (9)$$

Here, t is a time slot, c is a CP, CT is a set of CPs, T is a set of operating times, and PT is a set of power service operations. Moreover, ε is a unit conversion variable (kW to kWh); τ_c^{ps} and τ_c^{pe} are the start and end times of the power service, respectively; τ_c^{ar} is the arrival time of the EV at the CP; η^c and η^d are the charging and discharging efficiency, respectively; $s_{t,c}^{min}$ and $s_{t,c}^{max}$ represent the minimum and maximum battery SOC; s_c^{it} is the battery SOC; and $P_{t,c}^{evm}$ is the energy consumed when driving to a CP.

4.2. EV User Response Model of EV Agent

EV users decide independently to participate in power services based on their attitudes toward the services (AT_t), incentives (δ_t), charging pressure (λ_t), and time pressure (γ_t). The responsiveness of EV users to participate in power services is quantified as a utility. The utility σ_s^t is based on the elasticity of price (incentive and charging price)

and the charging amount [32]. Assuming that charging of the EV does not move to another time, the charging amount, including self-elasticity (E_t^{et}), with a negative value can be expressed as Equation (10), which is derived through Equations (11)–(13) [32–35]. Here, Equation (11) represents the benefit of participating in the power service for an EV; when the profit maximization condition is considered, Equation (11) can be expressed as Equation (12). Equation (13) represents a quadratic function for the customer benefit ($B(d_t)$); differentiating it and applying Equation (12) yields Equation (10).

$$d_t = d_t^0 \left[1 + E_t^{et} \frac{(\rho_t - \rho_t^0 - \delta_t)}{\rho_t^0} \right], E_t = \frac{\Delta d_t}{\Delta \rho_t} \leq 0 \quad (10)$$

$$F(d_t) = B(d_t) - d_t \cdot \rho_t + \delta_t \cdot \Delta d_t \quad (11)$$

$$\frac{\partial B(d_t)}{\partial d_t} = \rho_t - \delta_t \quad (12)$$

$$B(d_t) = B_t^0 + \rho_t^0 (d_t - d_t^0) \left[1 + \frac{d_t - d_t^0}{2E_t^{et} \cdot d_t^0} \right] \quad (13)$$

Here, d_t^0 is the initial charging energy, ρ_t^0 is the initial charging price, E_t^{et} is self-elasticity, ρ_t is the charging price, and δ_t is the incentive.

If users' time pressure (γ_t) and charging pressure (λ_t) are applied to the elasticity model, it can be expressed as a utility for individual EV users, as in Equation (14) [32]. Here, time pressure is defined as the EV driver's spare time, including the charging time and the driving time; more spare time leads to a higher possibility of participating in the power service. Charging pressure is expressed as the degree of battery SOC required for EV driving; the lower the SOC, the higher the possibility of participating in the power service. Time pressure can be expressed based on the user's time, as in Equation (15), while charging pressure can be expressed based on the remaining battery capacity, as in Equation (16) [32,36,37].

$$\sigma_t = d_t^0 \left[1 + \frac{E_t^{et}}{\rho_t^0} (\rho_t - \rho_t^0 - \frac{\gamma_t \cdot \delta_t}{\lambda_t} \cdot \omega) \right] \quad (14)$$

$$\gamma_t = \left(\frac{\min(t, \tau_t^{ts})}{\tau_t^{ts}} \right)^{\frac{1}{AT_t}} \quad (15)$$

$$\lambda_t = \left(\frac{B_t^S}{B^c} \right)^\alpha \quad (16)$$

Here, ω represents the weight variable, AT_t represents the EV user's attitude, α denotes the charging pressure parameter, and B_t^S and B^c represent the battery SOC and battery capacity, respectively.

4.3. Power Service Optimization Model of CPO

In this study, the power service was an incentive-based DR program aimed at reducing renewable energy curtailment. The operational goal of the CPO is to determine the optimal incentive to induce EV charging. When the DR signal is announced, the CPO adjusts incentives to increase the charging capacity of the CP to induce EV charging participation. The CPO operates with the goal of maximizing the operation profit, which is calculated based on the DR response power, charging price, and incentives, using Equation (17) [32]. Equation (18) represents the DR response power, which is calculated as the sum of the EV charging amount during the DR operation time. Here, the EV charging is assumed to be the rated charging. Equations (19) and (20) represent the incentive constraints and the DR contract capacity, respectively.

$$\text{maximize } \sum_{t \in PT} \sum_{c \in CT} P_{t,c}^{ps} (\rho_{t,c}^0 - \delta_{t,c}) \varepsilon \quad (17)$$

$$P_{t,c}^{ps} = \sum_{i \in EV} u_{t,c,i}^* P_{c,i}^{cp}, \forall t \in PT, \forall c \in CT, \forall i \in EV \quad (18)$$

$$\delta_{t,c} \leq \delta_t^{max}, \forall t \in PT, \forall c \in CT \quad (19)$$

$$\varepsilon P_{t,c}^{ps} \geq P_{t,c}^{con}, \forall t \in PT, \forall c \in CT, \forall i \in EV \quad (20)$$

Here, i denotes an EV, EV is the EV set, $P_{t,c}^{ps}$ is the DR response power of the CPs, δ_t^{max} is the maximum incentive, and $P_{t,c}^{con}$ is the DR contract capacity.

The CPO's optimization module calculates the optimal incentive by interacting with the EV agent's state module. The incentive for the CPO is reflected as an input to the EV response module linked to the EV state module. The utility calculated in the EV response module is reflected in the EV state module, and the results of the EV user's CP selection and charging amount are reflected as inputs to the CPO's optimization module. The particle swarm optimization model was used to derive the optimal result.

5. Case Study and Discussion

5.1. Case Study Design

A case study targeting 120 EVs was conducted considering the actual EV charging pattern with respect to time. The distribution system was a simplified version of the actual distribution system in Jeju Island, South Korea, and approximately 6.5 MWp of PV was assumed to have been introduced. Additionally, three CPs were placed at 5 km intervals. The rated power of the CP charger was set to 50 kW, considering fast charging. The power service was targeted at incentive-based DR to manage overvoltage in the distribution system. The DR information (operating times, contract information, etc.) was announced to the CPO 15 min before the DR started, and the DR operation time and DR contract capacity were set to 30 min and 1.44 MWh, respectively. In addition, the charging price is 18 ¢/kWh, and the maximum incentive was set at 54% of the charging price. The self-elasticity of the EV reactivity model was set to -1 considering the rated charger. The AT and α were set to 0.9–1.1 and 0.1–0.3, respectively. Simulations were conducted at 1 min intervals, with analyses based on the DR operational times. Figure 6 shows the target distribution system.



Figure 6. Test distribution system.

5.2. Case Study Results and Discussion

(A) DR Program Operation Results

Figure 7 shows the CPO's optimization results. The left-side chart in Figure 7 presents the optimization progress for determining the CPO's profit and incentives for each CP. The right-side chart presents the determined incentives of each CP. According to the DR operation, the optimal incentives for CPs 1–3 were 4.46, 3.56, and 3.60 ¢/kWh, respectively. The optimal incentives converged after 29 iterations of the optimization progress, and the CPO's profit was approximately \$90.

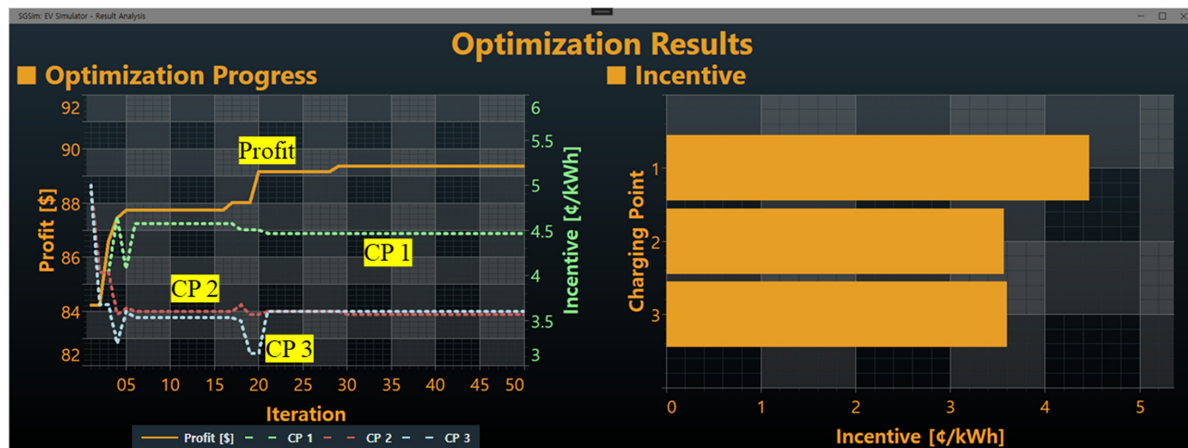


Figure 7. Optimization results for CPO.

In the DR service, 68 out of 120 EVs participated, among which 30 EVs had no prior charging plans. However, they were determined to participate in DR due to utility increases through incentives. The average charging power for DR participation across the three CPs was 1.478 MWh, resulting in an approximate 33.7% increase in profit compared to the charging revenue of CPO. These results indicate that incentive-based operation can encourage EV users to participate in DR, which increases CP utilization and, consequently, enhances CPO profit. Figure 8 shows the state information of the EVs before and after DR participation.

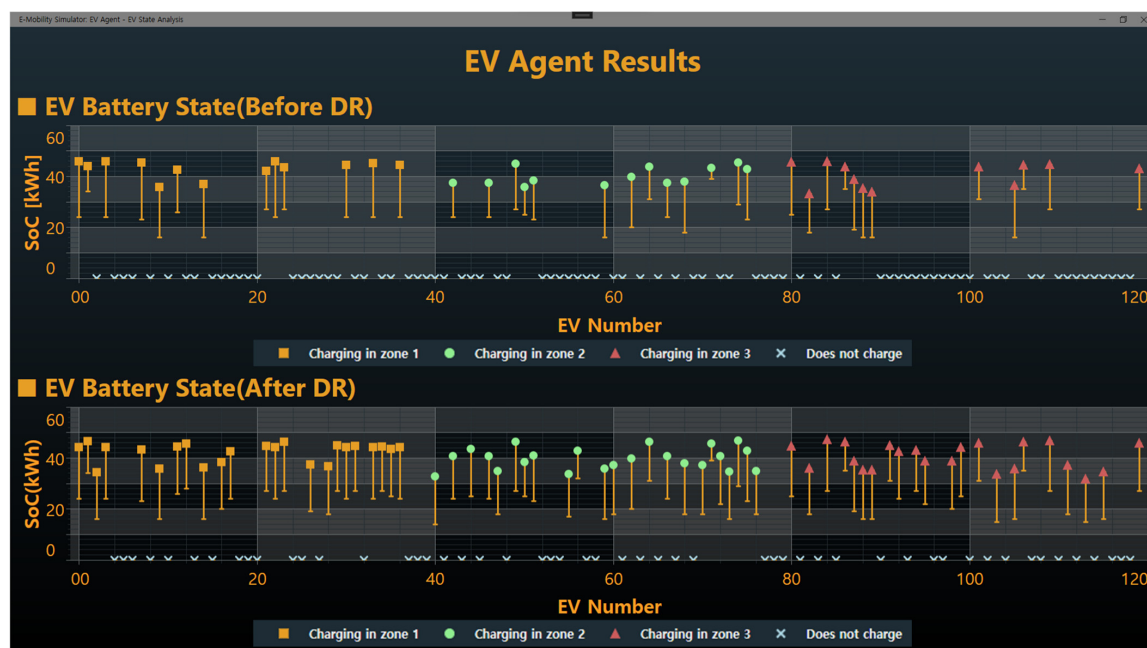


Figure 8. EV battery states before and after DR.

The EV user selects the CP that maximizes utility, considering factors such as distance to the CP (to calculate battery consumption), charging price, incentives, spare time, and battery state. The behaviors of individual EV users are illustrated in Figure 9, where “moving” represents the travel time to the CP, “staying” represents the waiting time, and “charging” represents the charging duration. Due to the incentives provided, the utility of the EVs participating in DR increased by an average of 10.04%. The utility and behavior results for each EV user are shown in Figure 9. The left-side chart in Figure 10 shows the voltage trend of the distribution system before and after DR participation. Overvoltage of up to 1.06 PU occurred around the terminal node (ID 21) before DR, and it was reduced to below the voltage management limit of 1.04 PU after DR. The right-side chart in Figure 10 presents the voltages of all nodes in the distribution system at a specific time, showing that the overvoltage in the distribution system was resolved after DR.

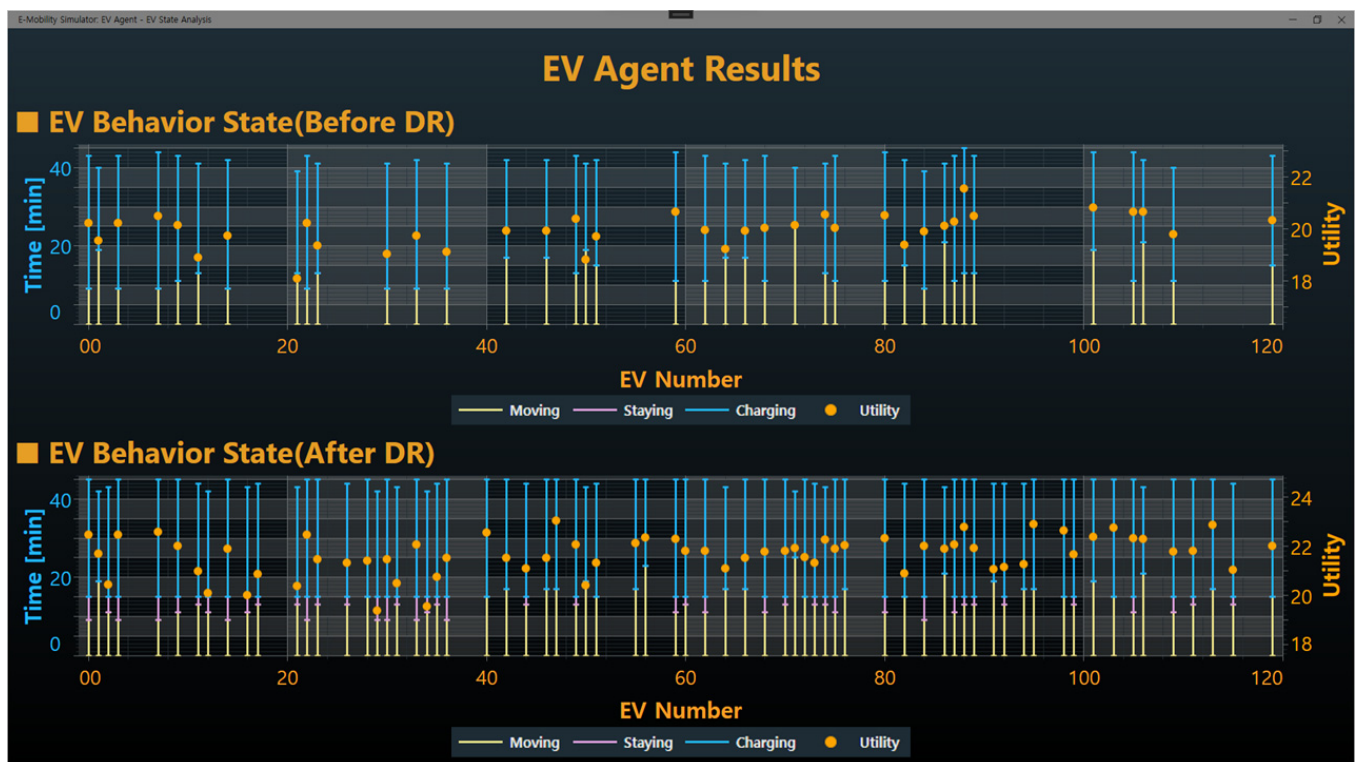


Figure 9. EV user behavior states before and after DR.

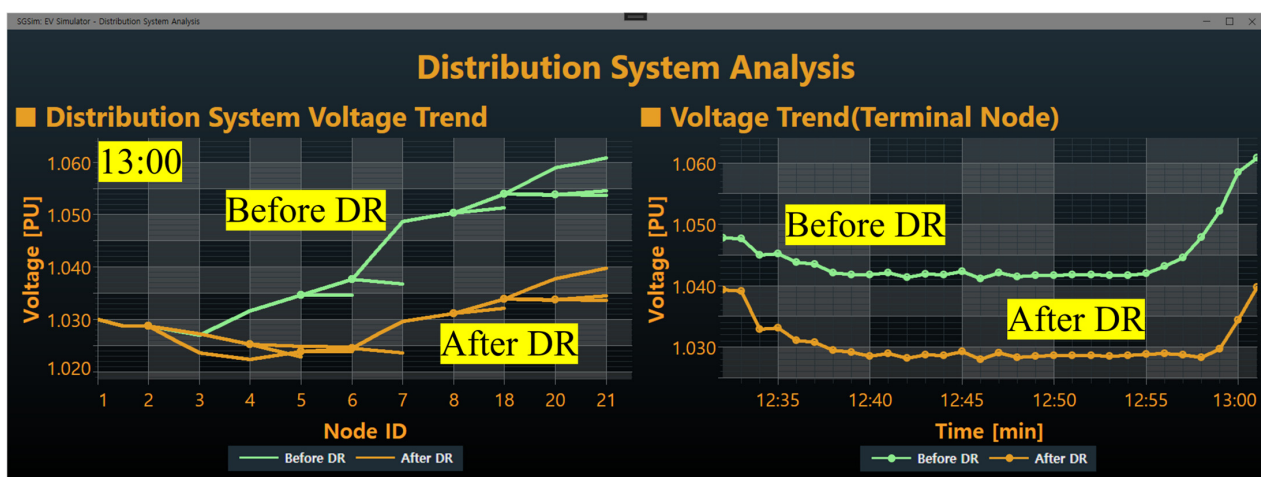


Figure 10. Distribution system voltage trends.

(B) Sensitivity Analysis for EV Users

To analyze the sensitivity of utility and CPO profit to the attitude of EV users toward DR participation, the parameter AT was varied by 50% of its initial value. A total of 300 Monte Carlo simulations were performed, the results of which are presented in Figure 11. The sensitivity analysis results show that the utility was distributed between 18.3 and 20.3, depending on the AT change of EV users. Depending on the responsiveness of these EV users, the CPO's profit varied in the range of \$74–103. The utility increased when the AT of EV users toward DR was positive, which was accompanied by an increase in DR participation and CPO profits. The characteristics of EV users can be adjusted to account for the simulation environment.

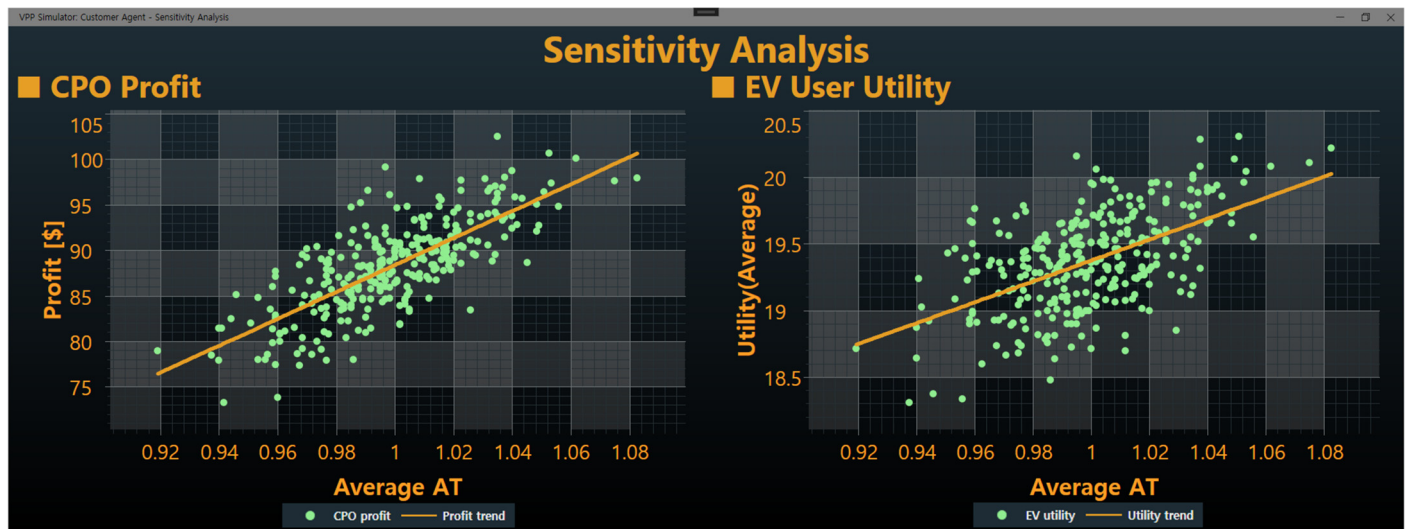


Figure 11. Sensitivity of CPO profits and EV user utility to EV user's attitude (AT) toward DR participation.

(C) Expandability and Interoperability

As shown in Figure 2, the agents interact over a databus-based architecture that is part of the SGSim platform. The architecture's message broker functionality enables essential data sharing to ensure seamless communication and minimize system delays while sharing relatively slow and non-critical data for analysis separately. The operation times of non-time-critical agents, such as analysis agents and core agents, can differ. Non-time-critical agents do not require synchronization with the core agents by allowing best-effort-based approaches. Owing to the proposed architecture, the system framework enables expandability to large-scale problems.

6. Conclusions

This study introduced a DT-based framework for simulating EV system operation, which can be linked with the simulation of power system operation. The proposed framework implements individual EVs, CPs, and CPOs as DTs to enable integrated operation with the power grid. The EV agent is implemented based on an advanced elasticity model that makes individual decisions and supports the behavioral simulation of EV users. These DT systems can be used to analyze the operational effects and grid impacts of power services based on their interactions with CPs and CPOs. The effectiveness of the proposed framework was verified via a case study on incentive-based DR operation. The case study results show that the framework can support the mutual operation and verification of EV users, CPOs, and grids.

The EV agent of the EV package can reflect regional and environmental characteristics through parameter adjustment. To support realistic and diverse simulations, further research will be conducted on deriving appropriate parameter values and incorporating

EV driving characteristics. Additionally, because the mobility of EVs affects user behavior, research on a response model for EV users that considers mobility characteristics will be conducted. In the future, EV behavioral response models will be developed as learning-based models, such as reinforcement and machine learning technologies, supporting more realistic EV-DT systems. Finally, an integrated framework that includes distribution system operators and market mechanisms will be established to simulate the operation of ancillary services under various distribution systems.

Author Contributions: Conceptualization, S.-W.P. and S.-Y.S.; methodology, K.-S.C., J.-U.Y. and S.-W.P.; software, K.-S.C.; validation, S.-W.P. and S.-Y.S.; formal analysis, J.-U.Y. and K.-S.C.; investigation, J.-U.Y. and K.-S.C.; resources, J.-U.Y. and K.-S.C.; data curation, K.-S.C.; writing—original draft preparation, J.-U.Y. and K.-S.C.; writing—review and editing, S.-W.P. and S.-Y.S.; visualization, K.-S.C.; supervision, S.-Y.S.; project administration, S.-Y.S.; funding acquisition, S.-Y.S. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Ministry of Trade, Industry and Energy (Grant No. RS-2024-00419642) supervised by the KETEP (Korea Institute of Energy Technology Evaluation and Planning) and the Gachon University research fund of 2020 (GCU-202004430001).

Data Availability Statement: Data are contained within the article. Further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare no conflicts of interest.

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